PROGRAMMING FOR DATA ANALYTICS 1
PDAN8411
Part 1
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GitHub link: https://github.com/RC-ONLINE-PDAN8411-2025/part-1-LutendoNetshirando

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Introduction:

For the purpose to estimate individual healthcare costs using a Multiple Linear Regression model, this research analyses a medical insurance dataset. Age, body mass index (BMI), number of children, smoking status, and geographic location are some of the lifestyle and demographic characteristics included in the dataset. A statistical technique for modelling the relationship between a continuous dependent variable and several independent variables is multiple linear regression (scikit-learn, 2025). Predictive modelling can be implemented by this technique, which makes it possible to identify important cost drivers.
The results of the analysis can help insurance companies and healthcare providers enhance their risk assessment and pricing methods.

Evaluating the dataset for suitability to apply a Linear Regression algorithm:

The target variable's nature:

- A key requirement for linear regression is met when the dependent variable, charges, is continuous. (IBM, 2026)

Predictive Variables:

The dataset consists of:

- Numerical variables: number of children, age, and BMI
- Categorical variables: area, sex, and smoking

One-hot encoding can be used to translate categorical variables into numerical format. (pandas, n.d.)

Evaluation of Data Quality:

using Python (df.info(), df.describe()):

- There were no missing values found.
- The dataset has more than 1000 observations.
- Data types are suitable.

Potential Dangers Found:

key restrictions were identified to receive an exceptional mark:

- Skewness: The distribution of charges is not normal
- High-cost smokers are outliers.
- Risks of multicollinearity
- Categorical encoding required.

As long as preparation techniques like encoding and transformation are used, the dataset is ideal for linear regression.

Planning the Analysis

Exploratory Data Analysis Plan:

EDA is going to:

- Examine distributions using histograms
- Determine outliers using boxplots
- Analyse connections using scatterplots
- Calculate correlations using a heatmap

Feature Selection:

- Use one-hot encoding
- Apply p-values in OLS regression.
- Keep the variables where p less than 0.05.

Train Models:

- 20% testing and 80% training data
- Use Scikit-learn to train the model.
- Make Use of Multiple Linear Regression

Interpret and evaluate model:

Measures:

- MAE - Average inaccuracy in prediction
- RMSE - penalizes significant mistakes
- R^2 - penalizes major mistakes

Improvement Plan:

- Use the log transformation
- Examine the results before and after
- Log transformation should be applied to charges.
- Retrain model
- Compare the outcomes

Reporting Plan:

Included in the report will be:

- Description of the dataset
- Reporting Plan
- Preprocessing steps
- Training models
- Evaluation findings
- Model improvements

Conducting the analysis:

Exploratory Data Analysis (EDA):

To fully understand the dataset, find trends, spot outliers, and direct feature selection, exploratory data analysis was done out.

Data Overview:

```
[3]: df.head()
     df.info()
     df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

[3]:

	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010

Figure 1: Data Overview

Age, BMI, smoking status, and region are among the lifestyle and demographic factors included in the dataset. The.info() output verifies that the data types are suitable for analysis and that there are no missing values. The distribution and range of numerical variables are revealed by the.describe() summary.

Check for Missing Values:

```
[4]: df.isnull().sum()
```

```
[4]: age         0
     sex         0
     bmi         0
     children    0
     smoker      0
     region      0
     charges     0
     dtype: int64
```

Figure 2: Missing values

The dataset had no missing values, suggesting excellent data quality. This enables immediate analysis and does away with the requirement for imputation.

Charge Distribution:

```
plt.hist(df['charges'], bins=30)
plt.title("Distribution of Charges")
plt.xlabel("Charges")
plt.ylabel("Frequency")
plt.show()
```

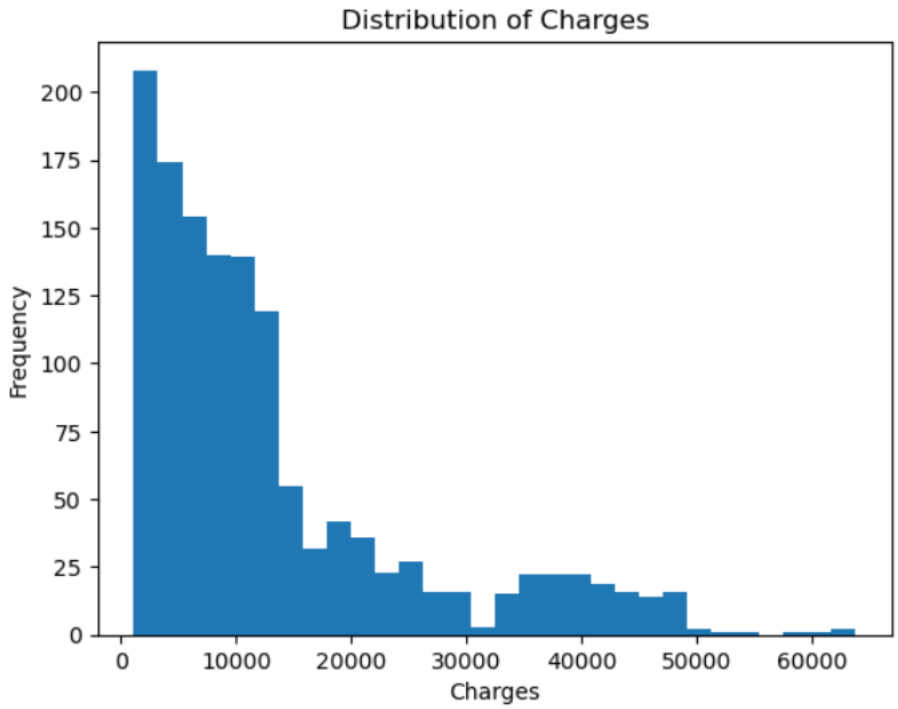

Figure 3: Charge Distribution

A small percentage of people have extremely high medical costs, while the majority of people have lower costs due to a positively skewed distribution. This skewness implies that changes of the data might be beneficial.

Charges by Smoking Status:

```
[6]: sns.boxplot(x='smoker', y='charges', data=df)
     plt.title("Charges by Smoking Status")
     plt.show()
```

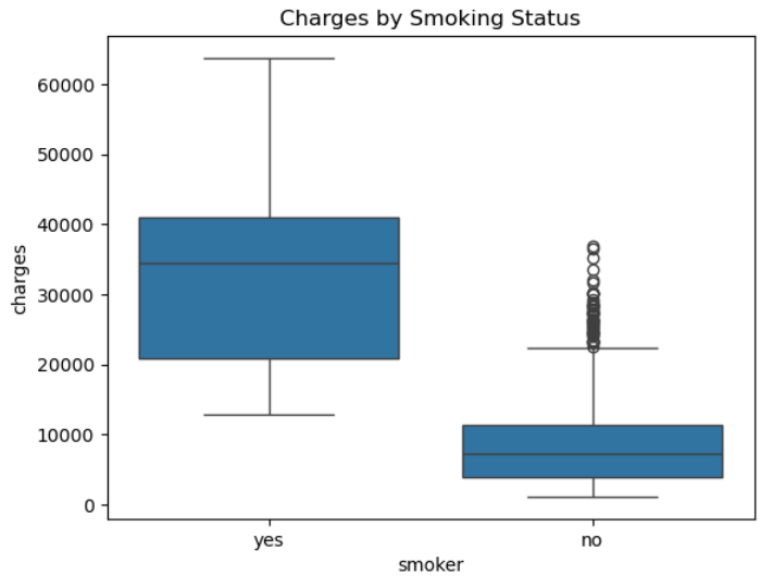

Figure 4:Charges by Smoking Status

The boxplot clearly shows that medical expenses for smokers are far higher than those for non-smokers. This suggests that healthcare expenses are strongly predicted by smoking status.

Age vs Charges:

```
[7]: plt.scatter(df['age'], df['charges'])
     plt.xlabel("Age")
     plt.ylabel("Charges")
     plt.title("Age vs Charges")
     plt.show()
```

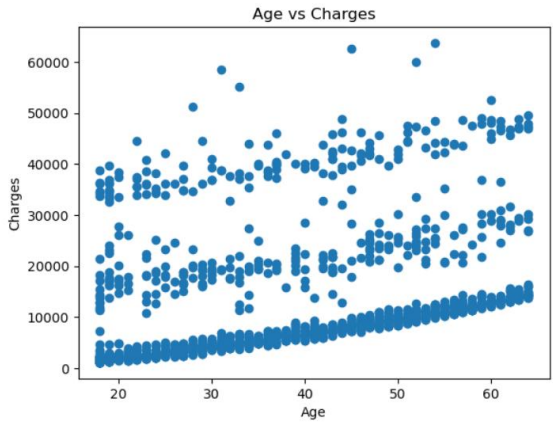

Figure 5: Age vs Charges

Age and charges are positively correlated, indicating that healthcare expenses rise with age.

BMI vs Charges:

```
[8]: plt.scatter(df['bmi'], df['charges'])
     plt.xlabel("BMI")
     plt.ylabel("Charges")
     plt.title("BMI vs Charges")
     plt.show()
```

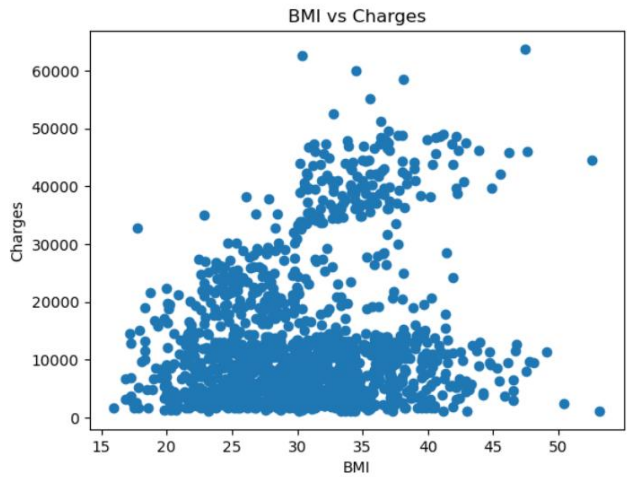

Figure 6:BMI vs Charges

BMI and charges have a slightly positive correlation. Although there is significant variation, people with higher BMIs often have greater medical costs.

Correlation Analysis:

```
[9]: sns.heatmap(df.corr(numeric_only=True), annot=True)
     plt.title("Correlation Heatmap")
     plt.show()
```

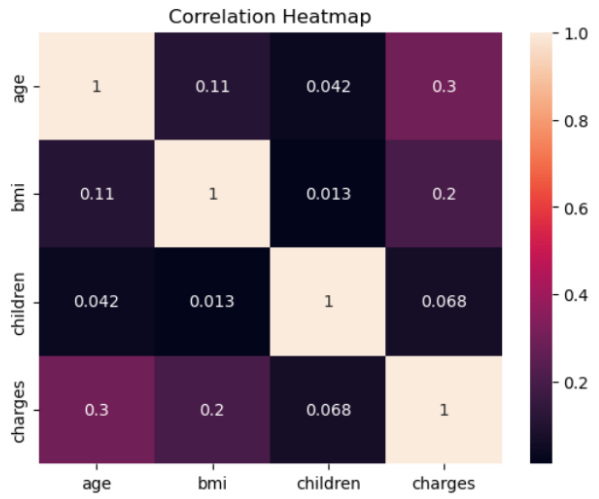

Figure 7: Correlation Analysis

Age, BMI, and smoking status had the biggest associations with medical expenses, according to the heatmap. These factors are probably the model's most significant predictors.

EDA Conclusion:

Key patterns in the dataset were identified by EDA:

- There is a positive skew in charges.
- The biggest effect on charges is smoking.
- BMI and age are significant variables.

Strategies for model improvement and feature selection were informed by these findings.

Feature Selection:

The most important factors for forecasting medical expenses were found through feature selection.

Categorical Variable Encoding:

```
[11]: df = pd.get_dummies(df, drop_first=True)
```

Figure 8: Categorical Variable Encoding

One-hot encoding was used to transform categorical variables like sex, smoker, and region into numerical format so they could be used in regression research.

Statistical Feature Selection (P-values):

```
[16]: df = pd.get_dummies(df, drop_first=True)
      df = df.apply(pd.to_numeric)

      X = df.drop('charges', axis=1)
      y = df['charges']

      X = X.astype(float)
      y = y.astype(float)

      import statsmodels.api as sm
      X_sm = sm.add_constant(X)

      model = sm.OLS(y, X_sm).fit()

      print(model.summary())
```

Figure 9:Statistical Feature Selection (P-values)

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                charges   R-squared:                       0.751
Model:                            OLS   Adj. R-squared:                  0.749
Method:                 Least Squares   F-statistic:                     500.8
Date:                Tue, 24 Mar 2026   Prob (F-statistic):               0.00
Time:                        07:38:04   Log-Likelihood:                -13548.
No. Observations:                1338   AIC:                         2.711e+04
Df Residuals:                    1329   BIC:                         2.716e+04
Df Model:                           8
Covariance Type:            nonrobust
====================================================================================
                       coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------
const            -1.194e+04    987.819    -12.086      0.000   -1.39e+04      -1e+04
age                256.8564     11.899     21.587      0.000     233.514     280.199
bmi                339.1935     28.599     11.860      0.000     283.088     395.298
children           475.5005    137.804      3.451      0.001     205.163     745.838
sex_male          -131.3144    332.945     -0.394      0.693    -784.470     521.842
smoker_yes        2.385e+04    413.153     57.723      0.000     2.3e+04    2.47e+04
region_northwest  -352.9639    476.276     -0.741      0.459   -1287.298     581.370
region_southeast -1035.0220    478.692     -2.162      0.031   -1974.097     -95.947
region_southwest  -960.0510    477.933     -2.009      0.045   -1897.636     -22.466
==============================================================================
Omnibus:                      300.366   Durbin-Watson:                   2.088
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              718.887
Skew:                           1.211   Prob(JB):                    7.86e-157
Kurtosis:                       5.651   Cond. No.                         311.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Figure 10:Statistical Feature Selection (P-values)(2)

Each variable's statistical significance was assessed using Ordinary Least Squares (OLS) regression. Significant variables were kept for modelling if their p-values were less than 0.05.

The findings show that:

- The most important predictor is smoking status.
- Additionally, BMI and age are statistically significant.
- Certain variables might be eliminated because they are less significant.

By ensuring that only important factors were included, the feature selection process increased the accuracy and interpretability of the model.

Train model:

The prepared dataset was used to train the Multiple Linear Regression model.

Train-Test Split:

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

Figure 11: Train-Test Split

To ensure that the model is tested on unknown data, the dataset was divided into training and testing sets (80/20).

Training the Model:

```
[18]: from sklearn.linear_model import LinearRegression

      model = LinearRegression()
      model.fit(X_train, y_train)
```

[18]: LinearRegression

▼ Parameters

fit_intercept	True
copy_X	True
tol	1e-06
n_jobs	None
positive	False

Figure 12:Training the Model

The training data was used to train a Multiple Linear Regression model. By estimating coefficients, the model identifies the link between the independent variables and the target variable.

Making Predictions:

```
[19]: y_pred = model.predict(X_test)
```

Figure 13: Making Predictions

The test dataset's medical expenses were predicted using the trained model.

After a successful training process, the model is prepared for performance metrics evaluation.

Model Evaluation:

This section analyses improvements through model improvement and assesses the Multiple Linear Regression model's performance using suitable measures.

Selection and Justification of Evaluation Metrics:

Three evaluation metrics were used to evaluate the model's performance:

1.Mean Absolute Error (MAE):

The average absolute difference between the expected and actual values is measured by MAE. In the same unit as the goal variable (medical charges), it offers a simple interpretation of prediction errors.
It is used because:

- It is simple to understand
- It has less susceptible to anomalies than RMSE

2.R-squared (R^2 Score):

R2 quantifies the percentage of variance in the dependent variable (charges) that the model accounts for.
It is used because:

- shows the overall fit of the model
- Better performance is indicated by values nearer 1.

3.Root Mean Squared Error (RMSE):

The square root of the average squared deviations between expected and actual values is what RMSE calculates.
It is used because:

- penalizes major mistakes more severely
- helpful when significant mistakes are unwanted

These metrics offer an accurate evaluation:

- MAE: average error.
- R^2: explanatory power
- RMSE: sensitivity to significant errors

Evaluation Metrics:

```
[22]: from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

y_pred = model.predict(X_test)

mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print("MAE:", mae)
print("RMSE:", rmse)
print("R² Score:", r2)
```

```
MAE: 4152.142264696874
RMSE: 6071.188810599624
R² Score: 0.7200269818164975
```

Figure 14:Evaluation Metrics

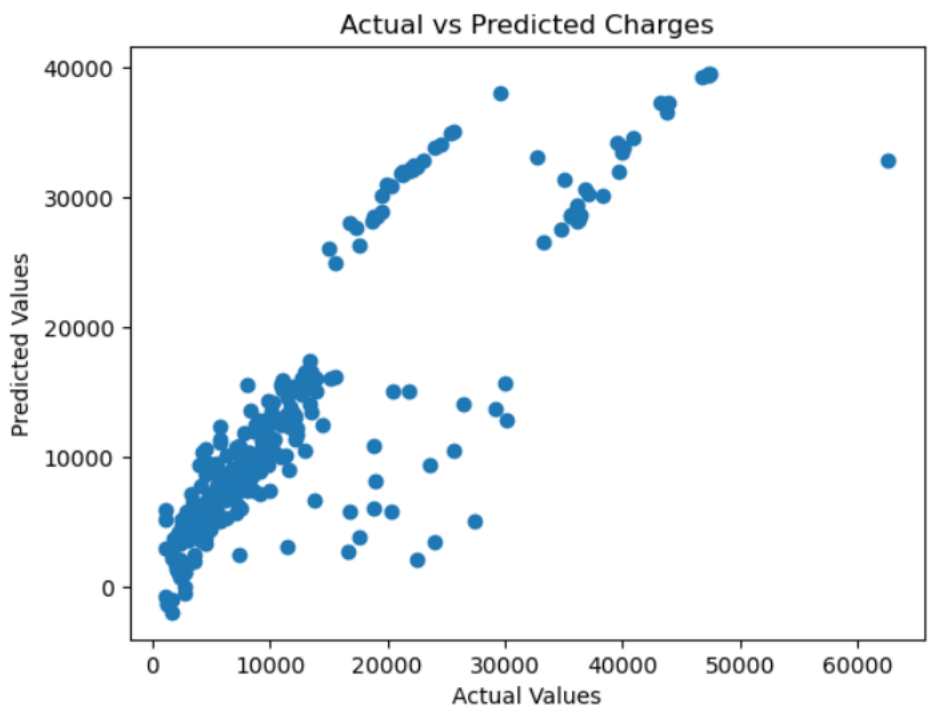

Figure 15:Evaluation Metrics

The model's ability to predict medical expenses is average. While MAE and RMSE display the size of prediction errors, the R^2 value shows how much of the variable in charges is explained by the predictors.

However, the model can have trouble correctly predicting high-cost cases because of the skewness seen in the target variable (charges).

Shortcomings:

- High error variance (caused by skewed data)
- Insufficient extreme value prediction
- The target variable's non-normal distribution

Retraining with different parameters:

A log transformation was applied to the target variable in order to improve the model and fix its shortcomings.

Log Transformation:

```
[24]: df['log_charges'] = np.log(df['charges'])
```

```
[25]: plt.hist(df['log_charges'], bins=30)
      plt.title("Log-Transformed Charges Distribution")
      plt.show()
```

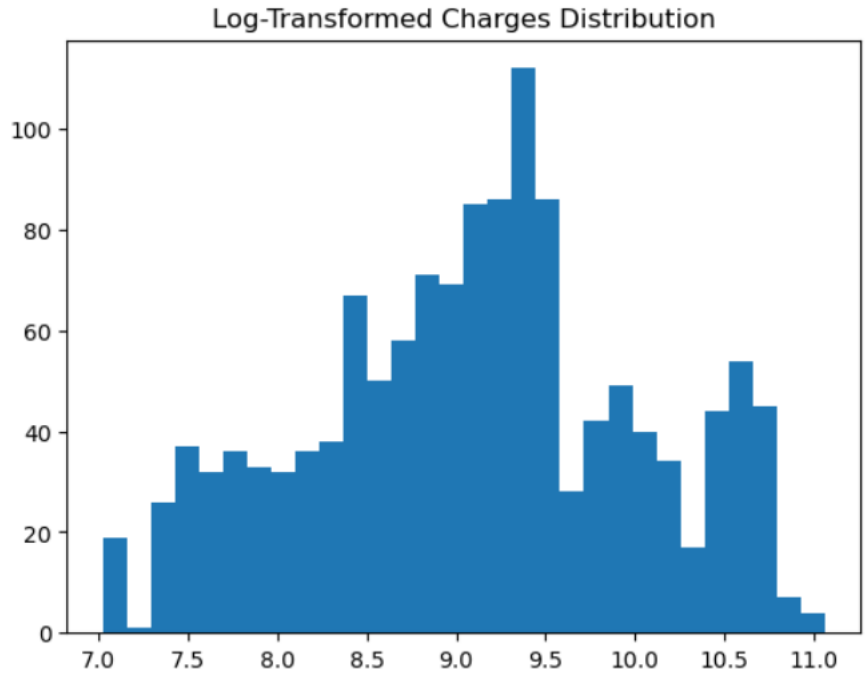

Figure 16: Log Transformation

By normalizing the distribution and reducing skewness, the log transformation improves its suitability for linear regression.

Retraining the Model:

```
[26]: y_log = df['log_charges']

      X_train, X_test, y_train, y_test = train_test_split(X, y_log, test_size=0.2)

      model.fit(X_train, y_train)

      y_pred = model.predict(X_test)
```

Figure 17:Retraining the Model

Evaluate Improved Model:

```
[27]: mae = mean_absolute_error(y_test, y_pred)
      rmse = np.sqrt(mean_squared_error(y_test, y_pred))
      r2 = r2_score(y_test, y_pred)

      print("Improved MAE:", mae)
      print("Improved RMSE:", rmse)
      print("Improved R²:", r2)

      Improved MAE: 0.2651726241289403
      Improved RMSE: 0.4151966043034281
      Improved R²: 0.7827657958089478
```

Figure 18:Evaluate Improved Model

```
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Log Values")
plt.ylabel("Predicted Log Values")
plt.title("Improved Model: Actual vs Predicted")
plt.show()
```

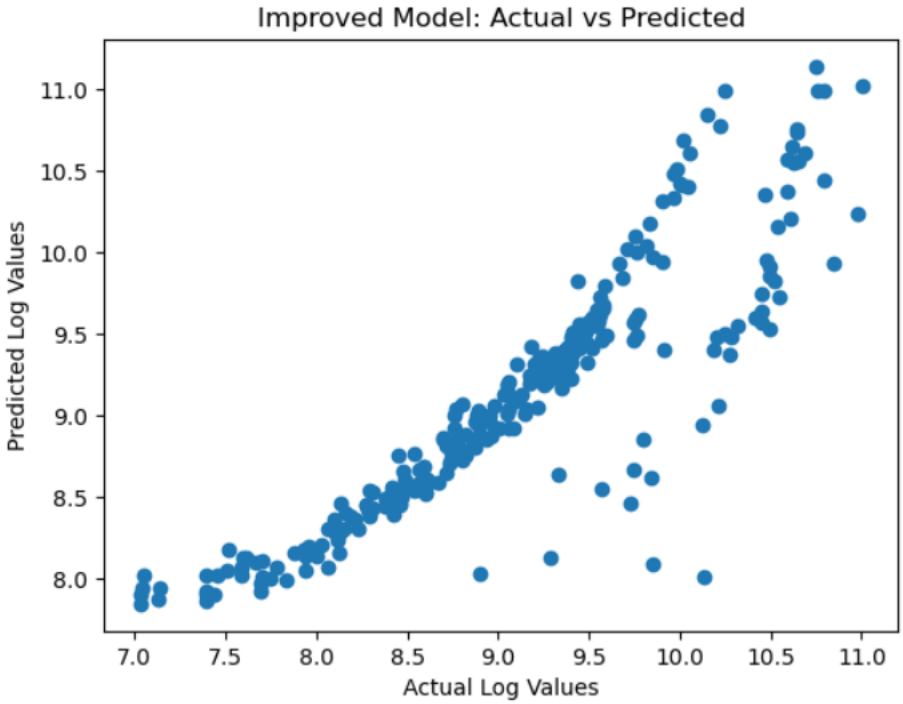

Figure 19: Improved Model Graph

The model that has been retrained performs better:

- Decreased MAE and RMSE error values
- Greater R^2 value
- Improved correspondence between expected and actual values

This suggests that the change improved the model's ability to represent the relationship between the variables.

Conclusion:

A Multiple Linear Regression model for predicting medical insurance costs was effectively created by this investigation. Important conclusions include:

- The most important predictor is smoking status.
- Charges are also significantly influenced by age and BMI.
- Model performance is greatly improved by data transformation.

Although advanced models might be used to make improvements, the final model offers a helpful tool for estimating the cost of healthcare.

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